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ACT SUBJECT TEST — MATH 2 • UNIT 5.01

Trigonometry, Identities & Polar Coordinates

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

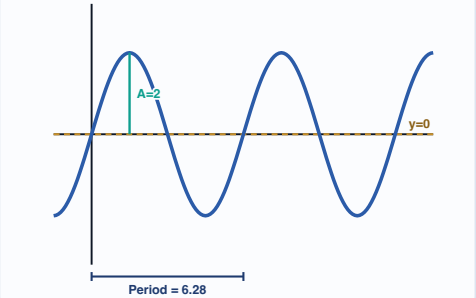
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 C	Q7 A	Q8 A	Q9 A	Q10 C
Q11 D	Q12 B	Q13 A	Q14 A	Q15 D
Q16 B	Q17 B	Q18 D	Q19 B	Q20 A
Q21 B	Q22 B	Q23 B	Q24 B	Q25 A

Trigonometry, Identities & Polar Coordinates Practice Exercises

1 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?



A $(-3, 3\sqrt{3})$ ✓

B $(-3\sqrt{3}, 3)$

C $(3, -3\sqrt{3})$

D $(3\sqrt{3}, -3)$

WORKED SOLUTION & VERIFICATION

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 6$ and $\theta = \frac{2\pi}{3}$:

$$x = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$$

$$y = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$$

Thus, the Cartesian coordinates are $(-3, 3\sqrt{3})$.

Common errors:

- Interchanging x and y (using $x = r\sin\theta$ and $y = r\cos\theta$) yields $(3\sqrt{3}, -3)$.
- Reversing the quadrant signs or applying $\frac{\pi}{6}$ reference angle formulas erroneously produces $(-3\sqrt{3}, 3)$.

2 MCQ

In $\triangle ABC$, the length of side AB is x, the length of side BC is $x+3$, and the length of side AC is 7. If $\angle B = 60^\circ$, what is the value of x?

- **A** 5 ✓
- **B** 8
- **C** 4
- **D** 6

WORKED SOLUTION & VERIFICATION

To find x, apply the Law of Cosines to $\triangle ABC$ with angle B:

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(B)$$

Substitute the given values $AC = 7$, $AB = x$, $BC = x+3$, and $\angle B = 60^\circ$:

$$7^2 = x^2 + (x+3)^2 - 2x(x+3)\cos(60^\circ)$$

Recall that $\cos(60^\circ) = \frac{1}{2}$:

$$49 = x^2 + (x^2 + 6x + 9) - 2x(x+3)\left(\frac{1}{2}\right)$$

$$49 = 2x^2 + 6x + 9 - x(x+3)$$

$$49 = 2x^2 + 6x + 9 - x^2 - 3x$$

$$49 = x^2 + 3x + 9$$

Rearrange into standard quadratic form:

$$x^2 + 3x - 40 = 0$$

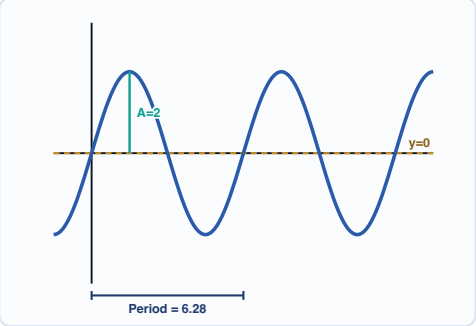
Factor the quadratic expression:

$$(x-5)(x+8) = 0$$

This yields $x = 5$ or $x = -8$. Since a triangle side length must be positive, we reject $x = -8$, giving $x = 5$.
Common traps:
- Solving for the length of side BC instead of x gives $x + 3 = 8$ (Choice B).
- Sign errors when expanding $(x-5)(x+8) = 0$ often lead students to select $x = 8$.

3 MCQ

In the polar coordinate plane, points A and B lie on the curve $r = 4\sin(2\theta)$ such that their polar angles satisfy $0 < \theta_A < \theta_B < \frac{\pi}{2}$ and $\theta_B - \theta_A = \frac{\pi}{3}$. What is the maximum possible area of $\triangle OAB$, where O is the origin?



A $\sqrt{3}$ ✓

B 2

C $2\sqrt{3}$

D $4\sqrt{3}$

WORKED SOLUTION & VERIFICATION

In polar coordinates, the area of a triangle formed by the pole O and two points $A(r_A, \theta_A)$ and $B(r_B, \theta_B)$ is given by:

$$\text{Area} = \frac{1}{2} r_A r_B \sin(\theta_B - \theta_A)$$

Given $\theta_B - \theta_A = \frac{\pi}{3}$, we have:

$$\text{Area} = \frac{1}{2} r_A r_B \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{4} r_A r_B$$

For points on the curve $r = 4\sin(2\theta)$, since $0 < \theta_A < \theta_B < \frac{\pi}{2}$, both $2\theta_A$ and $2\theta_B$ lie in $(0, \pi)$, so $r_A > 0$ and $r_B > 0$.
Let $\theta_A = \alpha$; then $\theta_B = \alpha + \frac{\pi}{3}$, where $0 < \alpha < \frac{\pi}{6}$.
Substituting r_A and r_B:

$$r_A = 4\sin(2\alpha), \quad r_B = 4\sin\left(2\alpha + \frac{2\pi}{3}\right)$$

$$\text{Area} = \frac{\sqrt{3}}{4} \cdot 16\sin(2\alpha)\sin\left(2\alpha + \frac{2\pi}{3}\right) = 4\sqrt{3}\sin(2\alpha)\sin\left(2\alpha + \frac{2\pi}{3}\right)$$

Using the product-to-sum identity $\sin(x)\sin(y) = \frac{1}{2}[\cos(x-y) - \cos(x+y)]$:

$$\sin(2\alpha)\sin\left(2\alpha + \frac{2\pi}{3}\right) = \frac{1}{2}\left[\cos\left(-\frac{2\pi}{3}\right) - \cos\left(4\alpha + \frac{2\pi}{3}\right)\right] = \frac{1}{2}\left[-\frac{1}{2} - \cos\left(4\alpha + \frac{2\pi}{3}\right)\right]$$

Since $0 < \alpha < \frac{\pi}{6}$, the angle $4\alpha + \frac{2\pi}{3}$ ranges across $\left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$. The cosine function reaches its minimum of -1 at $4\alpha + \frac{2\pi}{3} = \pi \implies 4\alpha = \frac{\pi}{3} \implies \alpha = \frac{\pi}{12}$, which lies inside the interval.
At this value, the product attains its maximum:

$$\frac{1}{2}\left[-\frac{1}{2} - (-1)\right] = \frac{1}{4}$$

Thus, the maximum possible area of $\triangle OAB$ is:

$$\text{Area}_{\max} = 4\sqrt{3} \cdot \frac{1}{4} = \sqrt{3}$$

A common error is to assume both r_A and r_B can independently reach the maximum curve radius of 4, which would incorrectly yield $\frac{\sqrt{3}}{4} \cdot 4 \cdot 4 = 4\sqrt{3}$. However, because $\theta_B - \theta_A = \frac{\pi}{3}$, the radii are coupled and cannot both equal 4 simultaneously.

4 MCQ

A point in the polar coordinate system has coordinates $(r, \theta) = (6, \frac{5\pi}{6})$. Which of the following represents the Cartesian coordinates (x, y) of this point?

A $(-3\sqrt{3}, 3)$ ✓

B $(-3, 3\sqrt{3})$

C $(3\sqrt{3}, -3)$

D $(-3\sqrt{3}, -3)$

WORKED SOLUTION & VERIFICATION

To convert polar coordinates (r, θ) to Cartesian coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 6$ and $\theta = \frac{5\pi}{6}$:

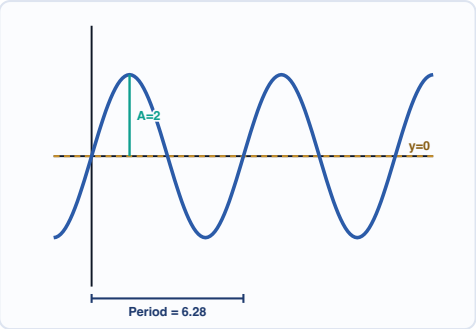
$$x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$$

$$y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$$

Thus, the Cartesian coordinates of the point are $(-3\sqrt{3}, 3)$. Common errors include swapping the cosine and sine components (resulting in $(-3, 3\sqrt{3})$) or making sign errors with Quadrant II trigonometric values.

5 MCQ

In $\triangle ABC$, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of $\angle B$ is 60°. What is the length of the altitude from vertex B to side AC?



A $\frac{20\sqrt{3}}{7}$ ✓

B $\frac{40\sqrt{3}}{7}$

C $\frac{10\sqrt{3}}{7}$

D $\frac{20}{7}$

WORKED SOLUTION & VERIFICATION

To find the altitude from vertex B to side AC, we first determine the length of side AC using the Law of Cosines:

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(\angle B)$$

Substitute the given values:

$$AC^2 = 5^2 + 8^2 - 2(5)(8)\cos(60^\circ) = 25 + 64 - 80\,(\frac{1}{2}) = 89 - 40 = 49$$

Taking the positive square root gives $AC = 7$.

Next, calculate the area of $\triangle ABC$ using the sine formula for triangle area:

$$\text{Area} = \frac{1}{2} \cdot AB \cdot BC \cdot \sin(\angle B) = \frac{1}{2} \cdot 5 \cdot 8 \cdot \sin(60^\circ) = 20 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3}$$

Finally, relate the area to the base AC and the altitude h from B:

$$\text{Area} = \frac{1}{2} \cdot AC \cdot h \implies 10\sqrt{3} = \frac{1}{2} \cdot 7 \cdot h \implies h = \frac{20\sqrt{3}}{7}$$

Common errors:

- $\frac{40\sqrt{3}}{7}$ results from omitting the factor of $\frac{1}{2}$ in the area formula.
- $\frac{10\sqrt{3}}{7}$ results from computing $\frac{\text{Area}}{AC}$ instead of $\frac{2 \cdot \text{Area}}{AC}$.
- $\frac{20}{7}$ results from mistakenly using $\cos(60^\circ) = \frac{1}{2}$ instead of $\sin(60^\circ) = \frac{\sqrt{3}}{2}$ when finding the area.

6 MCQ

In the polar coordinate system, point A has polar coordinates (r_A, θ_A) lying on the curve $r = 6\sin\theta$, and point B has polar coordinates (r_B, θ_B) lying on the curve $r = 8\cos\theta$, with $r_A > 0$ and $r_B > 0$. If $\theta_A - \theta_B = \frac{\pi}{6}$, what is the maximum possible area of $\triangle OAB$, where O is the pole (origin)?

A 6

B $6\sqrt{3}$

C 9 ✓

D 12

WORKED SOLUTION & VERIFICATION

In polar coordinates, the area of a triangle with one vertex at the pole $O(0,0)$ and the other two vertices at $A(r_A, \theta_A)$ and $B(r_B, \theta_B)$ is given by:

$$\text{Area}(\triangle OAB) = \frac{1}{2} r_A r_B \sin|\theta_A - \theta_B|$$

We are given $\theta_A - \theta_B = \frac{\pi}{6}$, so $\sin|\theta_A - \theta_B| = \sin(\frac{\pi}{6}) = \frac{1}{2}$.
Let $\theta_B = \theta$. Then $\theta_A = \theta + \frac{\pi}{6}$. Since point A lies on $r = 6\sin\theta$ and point B lies on $r = 8\cos\theta$, we have:

$$r_A = 6\sin(\theta + \frac{\pi}{6}), \quad r_B = 8\cos\theta$$

Substituting these expressions into the area formula:

$$\text{Area} = \frac{1}{2}[6\sin(\theta + \frac{\pi}{6})](8\cos\theta)(\frac{1}{2}) = 12\sin(\theta + \frac{\pi}{6})\cos\theta$$

Using the product-to-sum trigonometric identity $\sin u \cos v = \frac{1}{2}[\sin(u+v) + \sin(u-v)]$ with $u = \theta + \frac{\pi}{6}$ and $v = \theta$:

$$\sin(\theta + \frac{\pi}{6})\cos\theta = \frac{1}{2}[\sin(2\theta + \frac{\pi}{6}) + \sin(\frac{\pi}{6})] = \frac{1}{2}[\sin(2\theta + \frac{\pi}{6}) + \frac{1}{2}]$$

Thus, the area simplifies to:

$$\text{Area} = 12 \cdot \frac{1}{2}[\sin(2\theta + \frac{\pi}{6}) + \frac{1}{2}] = 6\sin(2\theta + \frac{\pi}{6}) + 3$$

The maximum value of $\sin(2\theta + \frac{\pi}{6})$ is 1, which occurs when $2\theta + \frac{\pi}{6} = \frac{\pi}{2} \implies \theta = \frac{\pi}{6}$.
Checking that $r_A, r_B > 0$ at $\theta = \frac{\pi}{6}$:
- $r_B = 8\cos(\frac{\pi}{6}) = 4\sqrt{3} > 0$
- $r_A = 6\sin(\frac{\pi}{6} + \frac{\pi}{6}) = 6\sin(\frac{\pi}{3}) = 3\sqrt{3} > 0$

Therefore, the maximum area is $6(1) + 3 = 9$.

7 MCQ

A point in the polar coordinate plane has coordinates $(r, \theta) = (4, \frac{5\pi}{6})$. Which of the following gives the rectangular coordinates (x, y) of this point?

- **A** $(-2\sqrt{3}, 2)$ ✓
- **B** $(-2, 2\sqrt{3})$
- **C** $(2, -2\sqrt{3})$
- **D** $(2\sqrt{3}, -2)$

WORKED SOLUTION & VERIFICATION

To convert polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 4$ and $\theta = \frac{5\pi}{6}$:

$$x = 4\cos\left(\frac{5\pi}{6}\right) = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$$

$$y = 4\sin\left(\frac{5\pi}{6}\right) = 4\left(\frac{1}{2}\right) = 2$$

Thus, the rectangular coordinates are $(-2\sqrt{3}, 2)$.

- Choice B is incorrect because it mistakenly swaps the values of $\cos\left(\frac{5\pi}{6}\right)$ and $\sin\left(\frac{5\pi}{6}\right)$.
- Choice C is incorrect because it swaps the x- and y-coordinates.
- Choice D is incorrect because it reverses the signs of both coordinates.

8 MCQ

In $\triangle ABC$, the side lengths are $AB = 5$ and $BC = 4$, and $\angle B$ is an acute angle. If $\cos(2B) = -\frac{31}{32}$, what is the length of side AC?

- **A** 6 ✓
- **B** $\sqrt{46}$
- **C** $\sqrt{41}$
- **D** $\sqrt{31}$

WORKED SOLUTION & VERIFICATION

We solve the problem in two steps by first determining $\cos(B)$ and then applying the Law of Cosines to find AC.

Step 1: Find $\cos(B)$

Using the double-angle identity for cosine:

$$\cos(2B) = 2\cos^2(B) - 1$$

Substitute the given value $\cos(2B) = -\frac{31}{32}$:

$$-\frac{31}{32} = 2\cos^2(B) - 1$$

$$2\cos^2(B) = 1 - \frac{31}{32} = \frac{1}{32}$$

$$\cos^2(B) = \frac{1}{64}$$

Since $\angle B$ is acute ($0^\circ < B < 90^\circ$), $\cos(B) > 0$:

$$\cos(B) = \frac{1}{8}$$

Step 2: Apply the Law of Cosines

In $\triangle ABC$, the side opposite $\angle B$ is AC:

$$AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos(B)$$

Substitute $AB = 5$, $BC = 4$, and $\cos(B) = \frac{1}{8}$:

$$AC^2 = 5^2 + 4^2 - 2(5)(4)\left(\frac{1}{8}\right)$$

$$AC^2 = 25 + 16 - \frac{40}{8} = 41 - 5 = 36$$

Taking the principal square root gives:

$$AC = 6$$

Common Traps:

- $\sqrt{46}$ arises from using an incorrect sign in the Law of Cosines ($41 + 5 = 46$) or erroneously setting $\cos(B) = -\frac{1}{8}$.
- $\sqrt{41}$ comes from assuming $\triangle ABC$ is a right triangle and applying the Pythagorean theorem directly ($\sqrt{5^2 + 4^2}$).
- $\sqrt{31}$ results from an arithmetic error such as computing $\cos(B) = \frac{1}{4}$ instead of $\frac{1}{8}$.

9 MCQ

In the polar coordinate plane, a curve is defined by the equation $r = 6\cos\theta + 8\sin\theta$. Two rays emanating from the pole at angles $\theta = \alpha$ and $\theta = \alpha + \frac{\pi}{2}$ intersect the curve at points $P \neq O$ and $Q \neq O$, respectively, where O denotes the pole. Assuming α is chosen such that both P and Q have positive radial coordinates, what is the maximum possible area of $\triangle OPQ$?

- **A** 25 ✓
- **B** 24
- **C** 48
- **D** 50

WORKED SOLUTION & VERIFICATION

We can solve this problem using either trigonometric synthesis or coordinate geometry.

Method 1: Trigonometric Synthesis

The radial coordinates of P and Q along their respective rays are:

$$r_1 = 6\cos\alpha + 8\sin\alpha$$

$$r_2 = 6\cos(\alpha + \frac{\pi}{2}) + 8\sin(\alpha + \frac{\pi}{2}) = -6\sin\alpha + 8\cos\alpha$$

Since the angle between the two rays is $(\alpha + \frac{\pi}{2}) - \alpha = \frac{\pi}{2}$, $\triangle OPQ$ is a right-angled triangle at the pole O. Its area is:

$$\text{Area} = \frac{1}{2}r_1 r_2 = \frac{1}{2}(6\cos\alpha + 8\sin\alpha)(8\cos\alpha - 6\sin\alpha)$$

Expanding the product:

$$(6\cos\alpha + 8\sin\alpha)(8\cos\alpha - 6\sin\alpha) = 48\cos^2\alpha - 36\sin\alpha\cos\alpha + 64\sin\alpha\cos\alpha - 48\sin^2\alpha$$

$$= 48(\cos^2\alpha - \sin^2\alpha) + 28\sin\alpha\cos\alpha = 48\cos(2\alpha) + 14\sin(2\alpha)$$

Multiplying by $\frac{1}{2}$:

$$\text{Area}(\alpha) = 24\cos(2\alpha) + 7\sin(2\alpha)$$

For any expression of the form $A\cos\phi + B\sin\phi$, the maximum value is $\sqrt{A^2 + B^2}$:

$$\text{Maximum Area} = \sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25$$

At this maximum, $\cos(2\alpha) = \frac{24}{25}$ and $\sin(2\alpha) = \frac{7}{25}$, which yields $r_1 = r_2 = 5\sqrt{2} > 0$, satisfying the conditions.

Method 2: Geometric Synthesis

Multiplying $r = 6\cos\theta + 8\sin\theta$ by r yields:

$$r^2 = 6r\cos\theta + 8r\sin\theta \implies x^2 + y^2 - 6x - 8y = 0 \implies (x-3)^2 + (y-4)^2 = 25$$

This is a circle with diameter $D = 2\sqrt{25} = 10$ passing through the pole $O(0,0)$. Because $\angle POQ = \frac{\pi}{2}$ is an inscribed angle in the circle, the hypotenuse PQ must be a diameter of the circle, so $PQ = 10$. The maximum area of any right triangle with hypotenuse c is $\frac{1}{4}c^2$:

$$\text{Area}_{\max} = \frac{1}{4}(10)^2 = 25$$

Misconception Analysis:

- Choice B (24) occurs if one mistakenly ignores the sine term and maximizes only $24\cos(2\alpha)$.
- Choice C (48) and Choice D (50) occur if one forgets the factor of $\frac{1}{2}$ in the triangle area formula $\frac{1}{2}ab$.

10 MCQ

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = (6, \frac{5\pi}{6})$?

A $(-3, 3\sqrt{3})$

B $(3\sqrt{3}, -3)$

C $(-3\sqrt{3}, 3)$ ✓

D $(-3\sqrt{3}, -3)$

WORKED SOLUTION & VERIFICATION

To convert from polar coordinates (r, θ) to rectangular coordinates (x, y), use the conversion formulas:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Given $r = 6$ and $\theta = \frac{5\pi}{6}$:

$$x = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$$

$$y = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$$

Thus, the rectangular coordinates are $(-3\sqrt{3}, 3)$.

- Choice A $(-3, 3\sqrt{3})$ results from inadvertently reversing the sine and cosine components (calculating $x = r\sin\theta$ and $y = r\cos\theta$).
- Choice C $(3\sqrt{3}, -3)$ results from applying Quadrant IV signs instead of Quadrant II signs.
- Choice D $(-3\sqrt{3}, -3)$ results from incorrectly treating both coordinates as negative (Quadrant III).

11 MCQ

Find $\sin(30^\circ)$.

A 1

B $-\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

$\sin(30^\circ) = \frac{1}{2}$ (reference-angle / unit-circle value).

12 MCQ

Find $\cos(180^\circ)$.

A 0

B **-1** ✓

C $-\frac{\sqrt{2}}{2}$

D $\frac{\sqrt{2}}{2}$

WORKED SOLUTION & VERIFICATION

$\cos(180^\circ) = -1$ (reference-angle / unit-circle value).

13 MCQ

Calculate $\cos(45^\circ)$.

- **A** $\frac{\sqrt{2}}{2}$ ✓
- **B** $\frac{1}{2}$
- **C** 1
- **D** $\frac{\sqrt{3}}{2}$

WORKED SOLUTION & VERIFICATION

$\cos(45^\circ) = \frac{\sqrt{2}}{2}$ (reference-angle / unit-circle value).

14 MCQ

Calculate $\sin(270^\circ)$.

- **A** -1 ✓
- **B** $\frac{\sqrt{3}}{2}$
- **C** 0
- **D** $\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

$\sin(270^\circ) = -1$ (reference-angle / unit-circle value).

15 MCQ

Compute $\cos(300^\circ)$.

- **A** 0
- **B** $\frac{\sqrt{2}}{2}$
- **C** $-\frac{1}{2}$
- **D** $\frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

$\cos(300^\circ) = \frac{1}{2}$ (reference-angle / unit-circle value).

16 MCQ

Compute $\sin(225^\circ)$.

- **A** 0
- **B** $-\frac{\sqrt{2}}{2}$ ✓
- **C** $-\frac{1}{2}$
- **D** $-\frac{\sqrt{3}}{2}$

WORKED SOLUTION & VERIFICATION

$\sin(225^\circ) = -\frac{\sqrt{2}}{2}$ (reference-angle / unit-circle value).

17 MCQ

Deduce $\sin(210^\circ)$.

- A 1
- B $-\frac{1}{2}$ ✓
- C $\frac{\sqrt{2}}{2}$
- D -1

WORKED SOLUTION & VERIFICATION

$\sin(210^\circ) = -\frac{1}{2}$ (reference-angle / unit-circle value).

18 MCQ

Obtain $\sin(180^\circ)$.

- A $\frac{1}{2}$
- B $-\frac{\sqrt{2}}{2}$
- C $\frac{\sqrt{2}}{2}$
- D 0 ✓

WORKED SOLUTION & VERIFICATION

$\sin(180^\circ) = 0$ (reference-angle / unit-circle value).

19 MCQ

Deduce $\cos(90^\circ)$.

- A $-\frac{1}{2}$
- B 0 ✓
- C -1
- D 1

WORKED SOLUTION & VERIFICATION

$\cos(90^\circ) = 0$ (reference-angle / unit-circle value).

20 MCQ

Obtain $\cos(120^\circ)$.

- A $-\frac{1}{2}$ ✓
- B $\frac{1}{2}$
- C -1
- D $-\frac{\sqrt{2}}{2}$

WORKED SOLUTION & VERIFICATION

$\cos(120^\circ) = -\frac{1}{2}$ (reference-angle / unit-circle value).

21 MCQ

Identify $\cos(30^\circ)$.

- A $-\frac{\sqrt{2}}{2}$
- B $\frac{\sqrt{3}}{2}$ ✓
- C $-\frac{1}{2}$
- D $-\frac{\sqrt{3}}{2}$

WORKED SOLUTION & VERIFICATION

$\cos(30^\circ) = \frac{\sqrt{3}}{2}$ (reference-angle / unit-circle value).

22 MCQ

Identify $\sin(90^\circ)$.

- A $-\frac{\sqrt{3}}{2}$
- B 1 ✓
- C $-\frac{1}{2}$
- D -1

WORKED SOLUTION & VERIFICATION

$\sin(90^\circ) = 1$ (reference-angle / unit-circle value).

23 MCQ

Establish $\sin(0^\circ)$.

- A $\frac{\sqrt{3}}{2}$
- B 0 ✓
- C -1
- D $-\frac{\sqrt{3}}{2}$

WORKED SOLUTION & VERIFICATION

$\sin(0^\circ) = 0$ (reference-angle / unit-circle value).

24 MCQ

Determine $\sin(300^\circ)$.

- A $\frac{\sqrt{3}}{2}$
- B $-\frac{\sqrt{3}}{2}$ ✓
- C $-\frac{\sqrt{2}}{2}$
- D 1

WORKED SOLUTION & VERIFICATION

$\sin(300^\circ) = -\frac{\sqrt{3}}{2}$ (reference-angle / unit-circle value).

25 MCQ

Establish $\cos(240^\circ)$.

- **A** $-\frac{1}{2}$ ✓
- **B** 1
- **C** -1
- **D** $\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

$\cos(240^\circ) = -\frac{1}{2}$ (reference-angle / unit-circle value).